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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	Claypot Chicken Rice
Selected Domain	Domain 1-Aiversecity
Selected SDG Track (1–17)	8
Project / Solution Name	Jobee
Name (Leader)	Lam Way Hou
Email & Phone (Leader)	lamwayhou@gmail.com 016-221-1236
University Affiliation(s)	Sunway University, Universiti Putra Malaysia
Number of Team Members	5

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

In today's rapidly evolving job market, both job seekers and employers face significant efficiency bottlenecks. Job seekers, particularly fresh graduates, often experience a "resume black hole",submitting hundreds of applications with little to no feedback, leading to high unemployment anxiety and skill mismatch. Conversely, employers are overwhelmed by massive volumes of unstructured resumes, resulting in prolonged hiring cycles and increased operational costs. According to industry benchmarks, recruiters spend an average of only 6-7 seconds scanning a single resume. This information asymmetry not only wastes valuable human capital but also exacerbates economic inefficiency and social anxiety. By leveraging artificial intelligence, we can bridge this gap, making the job-hunting process more transparent and reducing the friction in talent acquisition. Our project directly addresses this critical issue, aiming to democratize access to career opportunities and support the United Nations SDG 8 (Decent Work and Economic Growth).

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

We propose "Jobee," an AI-powered intelligent career matching platform. Jobee moves beyond traditional keyword matching by utilizing Large Language Models (LLMs) for deep resume parsing and role skill analysis. It accurately extracts users' skills and experiences, then compares them with the skill profiles in the O*NET occupational database. It recommends suitable tech roles, shows the key skills users are missing, and also considers their interests and the causes they care about. For each chosen role, Jobee generates a personalized learning roadmap with curated resources. It also sets Bonus Quests, small projects users submit as GitHub repositories, which are reviewed and scored with feedback. Our own forecasting models, trained on public data from Indeed Hiring Lab, Malaysia's DOSM, Hacker News, Epoch AI and METR, show how job demand, AI adoption and AI capability may change. Each forecast comes with its uncertainty range.

Building upon the technical skills gained from SDC Phase 1 workshops, we plan to integrate an OpenRouter-compatible API to access powerful Large Language Models (LLMs) for natural language processing. For secure and scalable data storage, we will utilize Supabase (PostgreSQL-based). Furthermore, we leverage cloud hosting platforms like Replit, Vercel and RunPod for seamless deployments.

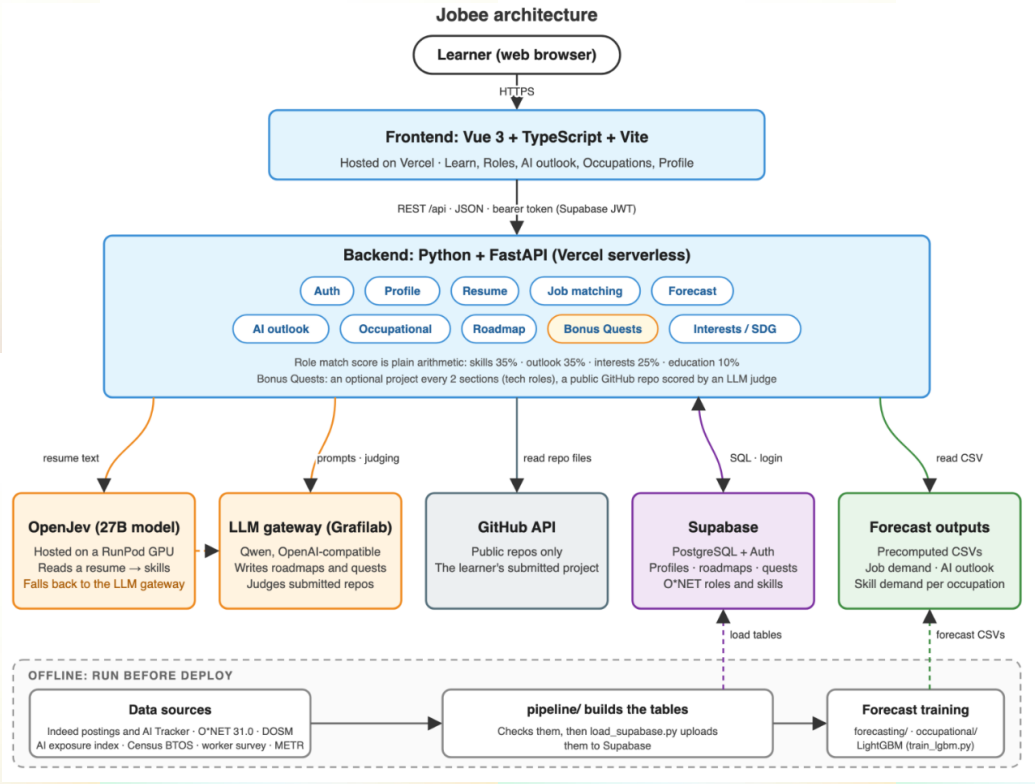
3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

- **Frontend:** Vue 3 + Vite + vue-router (TypeScript), hosted on Vercel. Pages: Learn, Roles, AI outlook, Occupations, Profile.
- **Backend:** Python + FastAPI on Vercel serverless functions. REST `/api` with JSON, authenticated by a Supabase JWT bearer token. Modules: Auth, Profile, Resume, Job matching, Forecast, AI outlook, Occupational, Roadmap, Bonus Quests, Interests / SDG.
- **Role matching:** A plain weighted score, not ML: skills 35%, AI outlook 35%, interests 25%, education 10%.
- **AI & GenAI:**
  - **OpenJev (27B model)** on a RunPod GPU. It extracts skills from resume text and falls back to the LLM gateway if it's unavailable.
  - **LLM gateway (Grafilab)**, OpenAI-compatible, serving Qwen. It writes roadmaps and Bonus Quests, and it acts as the LLM judge that scores submitted GitHub repos.
- **External APIs:** GitHub API, used to read files from public repos that learners submit for Bonus Quests.
- **Database & Auth:** Supabase. It provides PostgreSQL (profiles, roadmaps, quests, O*NET roles and skills) plus Supabase Auth for login and JWTs.
- **Machine learning / forecasting:** LightGBM (`train_lgbm.py` in `forecasting/` and `occupational/`). It's trained offline and outputs precomputed CSVs for job demand, AI outlook and skill demand per occupation, which the backend reads at runtime.
- **Data pipeline:** Python scripts in `pipeline/` that run offline before deploy. They validate the source data, and `load_supabase.py` uploads the tables to Supabase.
- **Data sources:** Indeed postings and AI Tracker, O*NET 31.0, DOSM, AI exposure index, Census BTOS, a worker survey, METR.
- **Cloud & deployment:** Vercel (frontend + serverless backend), RunPod (GPU for OpenJev), Supabase (managed database and auth).



3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

Our team applied a user-centric design-thinking methodology. We began by empathizing with job seekers facing application fatigue and recruiters overwhelmed by unstructured resumes. We defined the core problem as a semantic mismatch between candidate skills and job descriptions. During ideation, we conceptualized an AI-driven matching engine using LLMs via OpenRouter.

For prototyping, we built a Vue 3 frontend and a FastAPI backend, utilizing Supabase to ensure scalable data storage. We tested our prototype with sample resumes and iterated on the UI for simplicity.

Regarding our deployment strategy, we prioritized accessibility and technical viability. We deployed the application on Replit, ensuring judges and users can access the live prototype immediately without complex environment setups. By utilizing Supabase for our database and OpenRouter for AI inference, our architecture is fully cloud-native and highly scalable. This ensures our solution is not just a theoretical concept, but a technically viable, ready-to-deploy product for Phase 2.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

*The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess **Innovation and Creativity** as well as **Feasibility of build**. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	**Target 8.6:** By 2020, substantially reduce the proportion of youth not in employment, education or training.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	In 2024, the average job seeker submits 294 applications and spends 247 days to secure a single job offer. The recruitment side is similarly inefficient: 72% of recruiters spend less than 2 minutes reviewing a single resume, and it takes an average of 339–660 hours to screen resumes for a single non-senior position. This bidirectional information mismatch results in a massive waste of time and opportunities.
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	Skill demand prediction accuracy: 77.5% AI adoption prediction accuracy: 85% AI acceleration prediction: 74.1% (3 months), 68.9% (6 months), 54.5% (12 months)
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	Increase skill demand prediction accuracy to 80%+ Increase AI acceleration prediction to overall 70%+

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- ☑ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- ☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- ☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- ☑ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
FastAPI	Code	MIT
Pydantic Settings	Code	MIT
pandas	Code	BSD-3-Clause
NumPy	Code	BSD-3-Clause
scikit-learn	Code	BSD-3-Clause
LightGBM	Code	MIT
openpyxl	Code	MIT
pypdf	Code	BSD-3-Clause
PyYAML	Code	MIT
OpenAI Python SDK	Code	Apache-2.0
Supabase Python client	Code	MIT
Vue	Code	MIT

Vue Router	Code	MIT
Lucide (lucide-vue-next)	Code	ISC
Vite	Code	MIT
@vitejs/plugin-vue	Code	MIT
TypeScript	Code	Apache-2.0
vue-tsc	Code	MIT
Vitest	Code	MIT
@types/node	Code	MIT
Qwen3.7 Flash (via OpenRouter)	Model	Proprietary (API terms)
Indeed Job Postings Index	Dataset	CC-BY-4.0
Indeed AI Tracker	Dataset	CC-BY-4.0
Anthropic Economic Index	Dataset	CC-BY
O*NET 31.0 Database	Dataset	CC-BY-4.0
DOSM Job Market Insights (Malaysia)	Dataset	Not stated
Census Business Trends and Outlook Survey (BTOS)	Dataset	Public domain
Real-Time Population Survey: GenAI adoption (FRED)	Dataset	Not stated
Epoch AI Benchmark Data (incl. METR time horizons)	Dataset	CC-BY-4.0
Epoch AI Models Data	Dataset	CC-BY-4.0
METR Time Horizon 1.1 benchmark results	Dataset	Not stated
Hacker News "Who is hiring?" (Algolia API)	Dataset	Not stated
Stack Overflow Developer Survey 2023–2025	Dataset	ODbL
Adzuna Jobs API	Dataset	Proprietary (API terms)

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☑No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team’s knowledge and that the project will be original work created during the SDC 2026 hackathon period.



25/10/2026